NANO LAB

Gemini Nano On-Device Evaluation

Unaffiliated technical brief | Stock Pixel 10 Pro | Android 17 | Updated August 19, 2026

Published by Jason McArdle | Developed and documented with extensive AI assistance

Nano Lab is an Android-only Flutter test harness used to observe how the tested ML Kit GenAI capabilities behaved on physical Pixel hardware, including practical strengths, failures, and uses requiring human review.

The central finding: repeatability is not the same as correctness or completeness.

Executive summary

On a stock Pixel 10 Pro, Gemini Nano was genuinely useful for narrow, reviewable tasks. It performed especially well at deterministic date sorting, factual extraction, proofreading, professional rewriting, and draft-quality speech recognition. Proofreading was the strongest controlled result: every planted error was corrected, all facts were preserved, nothing was added, four outputs were identical, and average processing time was approximately 1.04 seconds.

Important limitations were also repeatable. Structured extraction added unwanted Markdown fences. Summarization was fast and factually sound but omitted the article's central results. Rewriting added an unrequested sign-off. Image Description repeatedly omitted a prominent tree and repeatedly misidentified an empty charging stand as a smartphone. Speech Recognition occasionally made meaningful substitutions.

STRONGEST RESULT Proofreading: 4 identical correct runs, 1.04 s average.	**KEY FAILURE MODE** Fluent and repeatable output could still be incomplete or wrong.
OFFLINE EVIDENCE All 6 capabilities worked after an offline restart once assets were installed.	**MEMORY OBSERVATION** Visible AICore PSS rose from about 209 MiB to 299 MiB after first prompt; no measured state approached 3 GB.

Full source, test documentation, recorded inputs and results, and report: github.com/mcardlejsn/nano-lab

1. Test design and environment

This was a focused, unaffiliated device evaluation, not a formal benchmark. Nano Lab exposes fixed inputs and relevant settings, then records outputs, processing times, completion state, and available finish information. Test content was fictional or disposable, and session history existed only in memory.

Item	Configuration
Device	Stock Google Pixel 10 Pro; 16 GB advertised RAM; locked bootloader
Operating system	Android 17; SDK 37 reported by the app
Application	Nano Lab; Android-only Flutter app; package com.mycarejournals.nano_lab
Native integration	Kotlin bridge through Flutter platform channels
Model platform	Google ML Kit GenAI APIs, Gemini Nano, and Android AICore
Prompt model	Stable model available; Preview model unavailable
Network design	No cloud Gemini API, analytics, account, database, or app-generated network requests
Test data	Fictional text and disposable fixed images; no production medical or caregiver data

Evaluation criteria

CORRECTNESS Factual accuracy, completeness, and unsupported additions.	**CONSISTENCY** Repeatability across identical controlled runs.
BEHAVIOR Instruction following, availability, downloads, and finish state.	**PRACTICALITY** Latency, offline operation, and suitability for real workflows.

APIs tested

Prompt 1.0.0-beta4; Summarization 1.0.0-beta1; Rewriting 1.0.0-beta1; Proofreading 1.0.0-beta1; Image Description 1.0.0-beta1; Speech Recognition 1.0.0-alpha1.

Methodological controls

- Fixed, visible inputs made identical reruns possible without silently changing content.

- Repeated runs were used where repeatability was central to the question.

- The controlled Top-K comparison held all recorded settings constant except Top-K, but used only one run per value.

- Speech timings were complete microphone-session times, including speaking and manual interaction, not pure inference latency.

- The offline pass used airplane mode plus explicitly disabled Wi-Fi; this is strong practical evidence, not packet-level network forensics.

Interpretation rule: a result was not treated as safe merely because it was fast, fluent, or identical across repeated runs.

2. Capability results

The table below condenses the controlled Pixel 10 Pro findings. Timings are representative of the tested inputs and should not be generalized to every device, prompt, image, speaker, locale, or future model update.

Capability	Observed result	Typical time	Important caution
Freeform Prompt	Flexible; deterministic sorting preserved all facts across 3 identical runs; structured extraction captured all supplied fields.	About 4-7 s	Undecorated JSON instruction was ignored; outputs required validation and sanitizing.
Summarization	Fast, identical, and factually sound across 3 runs.	1.91 s avg	Consistently omitted the article's central results; freeform prompting was slower but more informative.
Rewriting	Preserved all supplied facts and produced professional wording across 3 runs.	5.21 s avg	Added an unrequested closing and [Your Name] placeholder every time.
Proofreading	Corrected every planted error, preserved every fact, added nothing, and returned identical output across 4 runs.	1.04 s avg	One short controlled sentence; not evidence that every possible correction will be perfect.
Image Description	Returned identical descriptions across four runs for each fixed image.	About 3.8 s warm	Omitted a prominent tree; called an empty charging stand a smartphone and reduced a visible mouse to a small black object.
Advanced Speech Recognition	Usually preserved the complete meaning; handled dates, quantities, and time well.	17.86 s avg session	About 4 meaningful substitutions across roughly 108 reference words; informal accuracy estimate only.
Offline operation	All 6 capabilities remained available and completed after an offline restart once assets were installed.	Comparable in 1 pass	Initial or future asset management may still require AICore connectivity.

Controlled generation settings

Temperature changed variation; a fixed seed at temperature 1.0 produced identical output twice; multiple candidates produced distinct responses. In a controlled Top-K comparison using values 1, 3, and 10, all outputs followed the uppercase and three-sentence instructions, but wording changed. A larger Top-K did not reliably produce a visibly more varied or better answer.

Top-K controls the eligible next-token pool. It is not a quality, knowledge, or answer-count setting.

Cross-capability pattern

BEST FIT Narrow tasks with explicit inputs, objective outputs, and easy human review.	**WEAKEST FIT** Tasks where an omission, substitution, or confident visual error could cause harm.
VALIDATION Use deterministic schema checks before accepting generated structured data.	**HUMAN ROLE** Keep a person in control of sending, saving, or acting on generated content.

3. Concrete success and failure cases

The most useful findings came from comparing output quality with repeatability. Several APIs produced identical results, but identical did not always mean complete or correct.

Test	What happened	Interpretation
Proofreading	A sentence containing spelling, capitalization, agreement, and word-form errors was corrected perfectly in 4 identical runs.	Strong evidence for short, reviewable proofreading on this controlled input.
Date sorting	Five fictional ISO-dated records were sorted correctly, with every fact preserved, across 3 identical temperature-0 runs.	Promising for deterministic local transformation when results are easy to verify.
Structured extraction	All supplied facts and ordered keys were present; a missing room became null. Every response still arrived inside Markdown json fences.	The facts were reliable, but generated structure should never be parsed blindly.
Summarization	The dedicated API was about 2.7 times faster than freeform prompting but repeatedly prioritized routine rules over central outcomes.	Factual correctness does not guarantee good prioritization or completeness.
Rewriting	Professional wording preserved the facts but repeatedly added Thanks and [Your Name].	Suitable for a draft, not exact-template generation or automatic sending.
Synthetic image	Four identical descriptions correctly captured the house scene but omitted the prominent tree every time.	A description can be accurate yet materially incomplete.
Real photograph	Four identical descriptions called an empty charging stand a smartphone and did not explicitly identify the mouse.	The clearest confident false assertion in the evaluation; repetition reinforced the same error.
Speech Recognition	Most meaning was preserved, but runs included substitutions such as Three to they, labeled to labor, and fictional to fixed.	Useful for draft transcription, but exact or safety-critical transcription requires confirmation.

The strongest practical lesson: repeatability should increase confidence only after correctness and completeness have been independently verified.

External qualitative comparison

The same tabletop photograph was later submitted separately to Gemini 3.7 Flash in the Gemini web app. Flash correctly identified the empty charging stand and the mouse. This was a one-image cloud comparison outside Nano Lab, not part of the on-device timing or offline measurements.

4. Offline operation and memory observations

Offline restart procedure

After required assets were available, the device was placed in airplane mode, Wi-Fi was explicitly disabled, Nano Lab was fully stopped, and the app was reopened while still offline. One test for each of the six capabilities was then completed while the device remained offline.

Capability	Offline result
Freeform Prompt	Completed
Summarization	Completed in 1.89 seconds
Rewriting	AVAILABLE; completed in 5.49 seconds
Proofreading	AVAILABLE; completed in 1.13 seconds
Image Description	AVAILABLE; completed in 3.85 seconds
Advanced Speech Recognition	AVAILABLE; completed in 21.76 seconds with an essentially perfect transcription

Release APK permission evidence

A fresh 48.2 MB release APK built successfully after flutter analyze reported no issues and all Flutter tests passed. AAPT2 inspection showed RECORD_AUDIO, the AICore BIND_SERVICE permission, and an application-specific dynamic-receiver permission. It contained neither INTERNET nor ACCESS_NETWORK_STATE.

Together, the offline pass and absent network permissions provide strong practical and package-level evidence that the tested ML Kit/AICore path operated without an active internet connection and that Nano Lab itself did not request network access.

Exploratory AICore memory sequence

Stage	AICore PSS	AICore RSS	Nano Lab PSS
Before Nano Lab opened	208.7 MiB	435.8 MiB	-
App opened; 6 clients constructed	201.5 MiB	431.9 MiB	192.2 MiB
After Stable availability check	201.7 MiB	432.4 MiB	241.6 MiB
After first fixed prompt	298.9 MiB	531.9 MiB	253.1 MiB

In this one release-build run, constructing clients and checking Stable availability did not materially increase visible AICore process PSS. The first prompt increased AICore PSS by about 97 MiB. No measured state showed a visible AICore process footprint approaching 3 GB.

This does not rule out GPU, accelerator, kernel, firmware, boot-reserved, or other allocations that Android does not attribute to ordinary process PSS. It is an exploratory observation, not a complete memory-accounting proof.

5. Interpretation, limitations, and next step

APPROPRIATE USES	INAPPROPRIATE USES
Draft proofreading, suggested rewrites, optional summaries, draft transcription, local text organization, and supplemental image descriptions - all with review.	Sole authority for medical, legal, financial, safety, exact transcription, autonomous actions, or accessibility decisions where errors could cause harm.

Important limitations

- Most testing was performed on one stock Pixel 10 Pro; this is not yet a cross-device benchmark.
- The test set was deliberately small and controlled and should not be generalized to every prompt, image, accent, locale, device, or model update.
- The tested APIs were beta or alpha and may change; the Preview Prompt model was unavailable.
- Top-K used one run per setting, so it did not establish per-setting repeatability or a statistical trend.
- Speech accuracy was estimated informally; session timing included speaking and manual interaction.
- Airplane mode is practical evidence, not packet capture or formal operating-system network instrumentation.
- The memory experiment was one exploratory run and cannot observe allocations outside ordinary Android process accounting.

Planned Pixel 11 Pro XL addendum

The next phase will repeat matched tests on a Pixel 11 Pro XL using the same inputs and settings. It will compare feature availability, downloads, outputs, processing time, speech behavior, offline operation, and AICore PSS/RSS at the same four memory checkpoints.

Conclusion: Gemini Nano is practically useful for local, reviewable assistance, but its output must be validated according to the risk of the task.

Development and AI-assistance disclosure

Nano Lab was developed through extensive collaboration with ChatGPT. AI assistance was used to develop much of the evaluation plan and test design, implement the app, research the APIs, troubleshoot problems, interpret technical concepts, and draft the documentation. Jason McArdle ran the documented procedures on physical Pixel hardware, supplied the real-world photograph, collected and reviewed the observed outputs, identified problems during testing, directed revisions, and is responsible for publishing this report. The measurements and device behavior came from Nano Lab and Android diagnostic tools operating with the physical Pixel 10 Pro; they were not simulated or generated by AI.

Technical evidence and references

Complete project: github.com/mcardlejsn/nano-lab
Full findings: NANO_LAB_FINDINGS.md
Official APIs: developers.google.com/ml-kit/genai

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